

# **AWS Interview Questions & Answers for 3-Year Java Developer**

## ***1. What is Amazon EC2 and how have you used it in projects?***

Amazon EC2 (Elastic Compute Cloud) provides resizable compute capacity in the cloud. As a Java developer, you may deploy Spring Boot or microservices apps on EC2 instances. Key tasks include launching instances, configuring security groups, SSH access, and auto-scaling.

## ***2. Difference between Amazon S3 and EBS?***

S3 is object storage (unlimited, serverless, used for static files, backups, media). EBS is block storage (attached to EC2, used for OS, DB storage, transactional workloads). Java apps often integrate with S3 using AWS SDK for file uploads/downloads.

## ***3. What is IAM and why is it important?***

IAM (Identity and Access Management) controls access to AWS resources. It provides users, groups, roles, and policies. For Java apps, IAM roles are preferable to store credentials securely instead of hardcoding.

## ***4. How do you connect Java applications with AWS services?***

By using AWS SDK for Java. Example: AmazonS3Client for S3 operations, AmazonDynamoDB client for DB operations, and AmazonSQS client for messaging. Spring Cloud AWS also simplifies integration.

## ***5. What is AWS Lambda and have you used it with Java?***

AWS Lambda is a serverless compute service where code runs without provisioning servers. You can write functions in Java to process S3 events, API Gateway requests, or DynamoDB streams. It's cost-efficient for event-driven workloads.

## ***6. Difference between RDS and DynamoDB?***

RDS is a managed relational database service (supports MySQL, PostgreSQL, Oracle, etc.), best for structured relational data. DynamoDB is a NoSQL key-value/document DB, highly scalable and serverless. Java developers use JDBC for RDS and AWS SDK for DynamoDB.

## ***7. Explain VPC in AWS.***

VPC (Virtual Private Cloud) is an isolated network in AWS where you can launch resources like EC2. You configure subnets (public/private), route tables, and security groups. For Java apps, VPC ensures secure DB connectivity inside private subnets.

## ***8. What is Elastic Beanstalk and why is it useful for Java devs?***

Elastic Beanstalk automates deployment, scaling, and monitoring of applications. Java developers can simply upload a WAR/JAR file, and Beanstalk provisions EC2, load balancers, scaling, and health checks automatically.

### ***9. How do you handle messaging/queueing in AWS?***

By using SQS (Simple Queue Service) and SNS (Simple Notification Service). SQS is for decoupling microservices using queues. SNS is for pub/sub notifications. Java apps consume messages using AWS SDK or Spring Cloud AWS.

### ***10. How do you ensure cost optimization in AWS?***

Use auto-scaling groups, choose right instance types, use reserved/spot instances, leverage S3 lifecycle policies, and CloudWatch monitoring. For Java workloads, consider moving to serverless (Lambda, Fargate) for cost efficiency.